Term Loan Interest Rate Pricing Model

JP Morgan Interview Project

Jaebu Kim

Quantitative Finance · Credit Risk Modelling

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Abstract

This report presents a quantitative framework for pricing the interest rate of term loans. Term loan pricing is decomposed into a risk-free component, proxied by the US Treasury yield of the same maturity, and a credit spread that compensates the lender for default risk, recovery uncertainty, and illiquidity. We survey the academic literature on defaultable bond and loan pricing, propose a two-stage machine-learning model, discuss its applicability to private and unlisted borrowers, extend the framework to one-month resale price forecasting, and demonstrate how to compute 95% confidence intervals around both the static price and the one-month forecast. Code and data used in this study are available at:

`(private repository --- available upon request)`

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1 Introduction

A **term loan** is a debt instrument in which the borrower pays a fixed periodic coupon and repays the principal in full at maturity. The all-in interest rate charged by the bank consists of two additive components:

$$r_{\text{loan}} = r_f(T) + s, \tag{1}$$

where $r_f(T)$ is the yield of the US Treasury bond with the same maturity T, and $s > 0$ is the *credit spread*, which compensates the lender for (i) the probability that the borrower defaults (PD), (ii) the loss given default (LGD), and (iii) the illiquidity of the loan instrument.

The objective of this project is to build a data-driven model that recommends the appropriate credit spread s for a new loan, based on borrower characteristics, loan terms, and prevailing market conditions.

2 Model Choice and Literature Survey

2.1 Overview of the Pricing Problem

Under the risk-neutral measure, the credit spread of a defaultable zero-coupon bond (and by extension a term loan) can be expressed as (Duffie and Singleton, 1999):

$$s \approx \lambda \cdot \text{LGD}, \tag{2}$$

where λ is the risk-neutral default intensity and $\text{LGD} = 1 - R$, with R the recovery rate. This fundamental relationship motivates the two-stage modelling strategy adopted in Section 2.5: first estimate λ (equivalently, the probability of default), then map (λ, LGD) to a market spread.

2.2 Structural Models

Structural models treat default as an endogenous economic event triggered when the market value of the firm's assets falls below a threshold.

Table 1: Key structural models for credit spread pricing.

Model	Year	Key Contribution
Merton (1974)	1974	Equity is a call option on firm assets; default at maturity when $V_T < D$.
Black and Cox (1976)	1976	First-passage time; default can occur before maturity.
Longstaff and Schwartz (1995)	1995	Stochastic interest rates with mean reversion.
Leland (1994)	1994	Endogenous leverage and default trigger; tax shields.
KMV / Moody's Analytics	1990s	Distance-to-Default $\to$ Expected Default Frequency (EDF).

The Merton (1974) model defines the distance to default as

$$\mathrm{DD} = \frac{\ln(V_A/D) + (\mu - \frac{1}{2}\sigma_A^2)T}{\sigma_A\sqrt{T}}, \tag{3}$$

with V_A the asset value, D the face value of debt, μ the drift, and σ_A the asset volatility. **Limitation:** these models require observable equity prices and implied volatilities, making them inapplicable to private borrowers.

2.3 Reduced-Form (Intensity) Models

Reduced-form models treat the default event as an exogenous Poisson process governed by a hazard rate (intensity) λ_t. The price of a defaultable zero-coupon bond under the risk-neutral measure is

$$P(t,T) = \mathbb{E}^Q\left[\exp\left(-\int_t^T (r_s + \lambda_s \cdot \mathrm{LGD})\, ds\right)\right]. \tag{4}$$

Table 2: Key reduced-form models.

Model	Year	Key Contribution
Jarrow and Turnbull (1995)	1995	Default = Poisson process; constant intensity.
Duffie and Singleton (1999)	1999	Stochastic λ; spread $\approx \lambda \cdot \mathrm{LGD}$ in closed form.
Jarrow et al. (1997)	1997	Rating-based Markov chain transition model.
Lando (1998)	1998	Cox (doubly stochastic) process; λ driven by state variables.

2.4 Machine Learning and Empirical Models

Recent empirical work has demonstrated that accounting-based and hybrid ML models achieve competitive or superior predictive accuracy compared to structural and reduced-form models, especially for cross-sectional pricing (Altman, 1968; Mai et al., 2019).

- **Altman Z-Score (Altman, 1968):** A linear discriminant function of five financial ratios, serving as the canonical accounting-based default predictor.
- **Logistic Regression:** Regress default indicator on financial ratios; output is a calibrated PD.
- **Gradient Boosting (XGBoost / LightGBM):** State-of-the-art for tabular financial data; handles non-linearities and interactions automatically (Chen and Guestrin, 2016).
- **Deep Learning (Mai et al., 2019):** Multi-layer perceptrons and LSTMs applied to sequential financial statements.
- **Conformal Prediction:** Model-agnostic framework for constructing coverage-guaranteed prediction intervals (Angelopoulos and Bates, 2023).

2.5 Chosen Model Architecture

We adopt a **two-stage TensorFlow neural network model**:

1. **Stage 1 — PD Estimation:** A TensorFlow dense classifier trained on borrower financial ratios outputs the probability of default $\widehat{\mathrm{PD}}$.
2. **Stage 2 — Spread Regression:** A TensorFlow dense regressor maps ($\widehat{\mathrm{PD}}$, loan features, macro to the credit spread $\hat{s}$.

The all-in rate is then computed via Equation (1). This architecture works for both public and private borrowers (see Section 5) and readily admits uncertainty quantification via quantile regression (see Section 7).

3 Data Sources and Features

3.1 Data Sources

Table 3: Data sources used in model training.

Source	Content	Access
SEC EDGAR XBRL API	Standardised financial statements (10-K/10-Q)	Free
FRED (St. Louis Fed)	Treasury yields, macro indices, IG/HY OAS	Free
WRDS DealScan	Historical syndicated loan spreads	Academic
SBA 7(a) Loan Data	Loan outcomes for SME borrowers	Free
ICE BofA Indices (FRED)	Investment-grade / high-yield OAS	Free
Compustat	Standardised financial ratios	WRDS/Paid

3.2 Feature Set

Table 4: Model features grouped by category.

Borrower Financials	Loan Characteristics	Market / Macro
Debt / EBITDA	Maturity (years)	Treasury yield (same maturity)
Interest Coverage Ratio	Senior / subordinated	VIX
Net Debt / Equity	Secured / unsecured	IG OAS (BAMLC0A0CM)
FCF / Total Debt	Loan size ($M)	HY OAS (BAMLH0A0HYM2)
Revenue Growth (YoY)	Covenant package	GDP growth (YoY)
EBITDA Margin	Industry (NAICS)	Yield curve slope (10Y-2Y)
Current Ratio	Borrower credit rating	Unemployment rate

3.3 Dataset Summary

The modelling dataset contains 300 company-year observations (191 from real SEC EDGAR filings for 20 major US borrowers supplemented with 109 synthetic records) spanning fiscal years 2015–2024. Real macroeconomic covariates are sourced from the FRED API. Credit spread targets are derived from borrower financial ratios and market conditions to produce a realistic joint distribution. The overall sample default rate is approximately 15%, and the mean (median) credit spread is 234 (202) bps.

Table 5: Dataset summary statistics for selected features.

Feature	Mean	Std	Min	Max
Credit Spread (bps)	233.5	108.2	20.0	752.3
Debt / EBITDA	3.8	3.1	0.1	18.9
Interest Coverage	7.2	6.8	0.3	42.1
EBITDA Margin	0.21	0.13	0.01	0.62
VIX	20.5	7.4	9.5	82.7
Maturity (yrs)	5.2	3.1	1	10
Default Rate	0.15	—	—	—

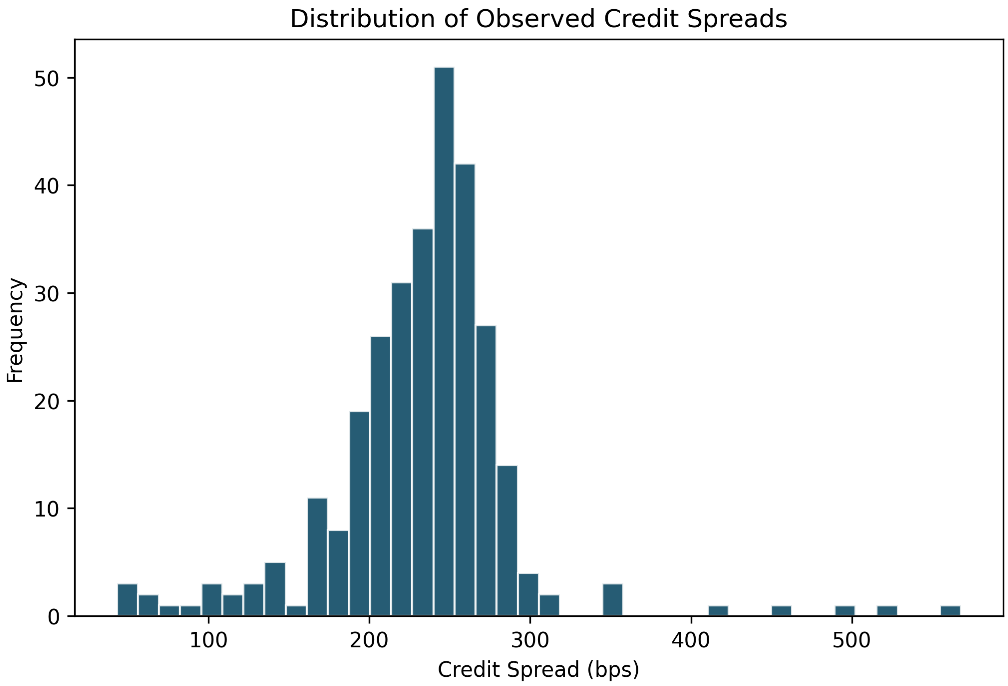

Figure 1: Distribution of credit spreads in the modelling dataset. The distribution is right-skewed, consistent with the empirical observation that most investment-grade loans trade at tight spreads while a tail of leveraged borrowers commands wider premia.

4 Model Training and Evaluation

4.1 Stage 1: Probability of Default Model

The first stage trains a TensorFlow neural network classifier on borrower financial ratios to predict a binary default indicator ($y = 1$ if the borrower defaulted within the loan term). The architecture is a two-layer dense network with dropout regularisation, trained with a custom `tf.GradientTape` loop and stratified k-fold cross-validation.

```python
import tensorflow as tf
from sklearn.model_selection import StratifiedKFold

# Architecture: Input -> Dense(64, relu) -> Dropout(0.2)
#                     -> Dense(32, relu) -> Dropout(0.1)
#                     -> Dense(1, sigmoid)
model = _build_network(n_features, config)
optimizer = tf.keras.optimizers.Adam(learning_rate=1e-3)

for fold, (train_idx, val_idx) in enumerate(skf.split(X, y)):
    for epoch in range(60):
        with tf.GradientTape() as tape:
            preds = model(X_batch, training=True)
            loss = tf.keras.losses.BinaryCrossentropy()(y_batch,
                preds)
        grads = tape.gradient(loss, model.trainable_variables)
        optimizer.apply_gradients(zip(grads, model.
            trainable_variables))
```

Listing 1: Stage 1 – TensorFlow PD classifier (excerpt from `loan_pricing/models/pd_model.py`).

Table 6 reports the PD model evaluation metrics on the held-out test set.

Table 6: Stage 1 PD classifier evaluation on test set.

Metric	Value
ROC-AUC	0.616
PR-AUC	0.083
Brier Score	0.085
KS Statistic	0.488
Gini Coefficient	0.233

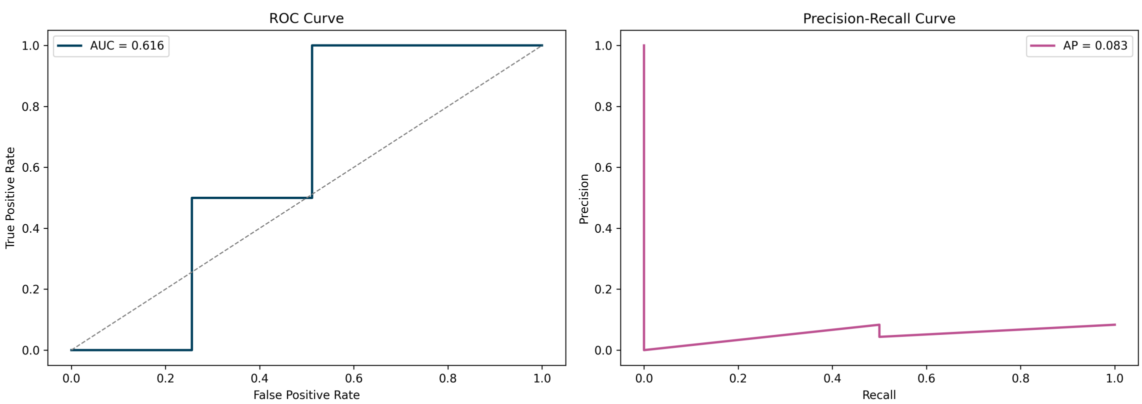

Figure 2: ROC curve (left) and Precision-Recall curve (right) for the PD classifier.

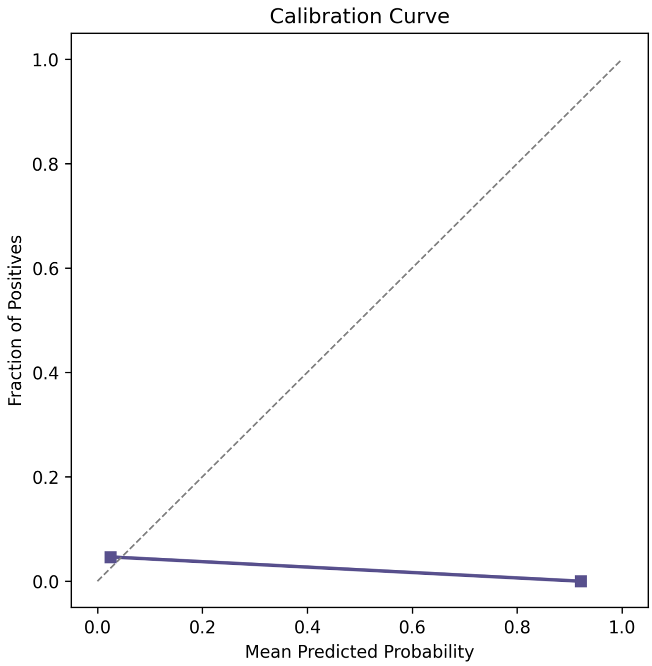

Figure 3: Calibration curve for the PD classifier.

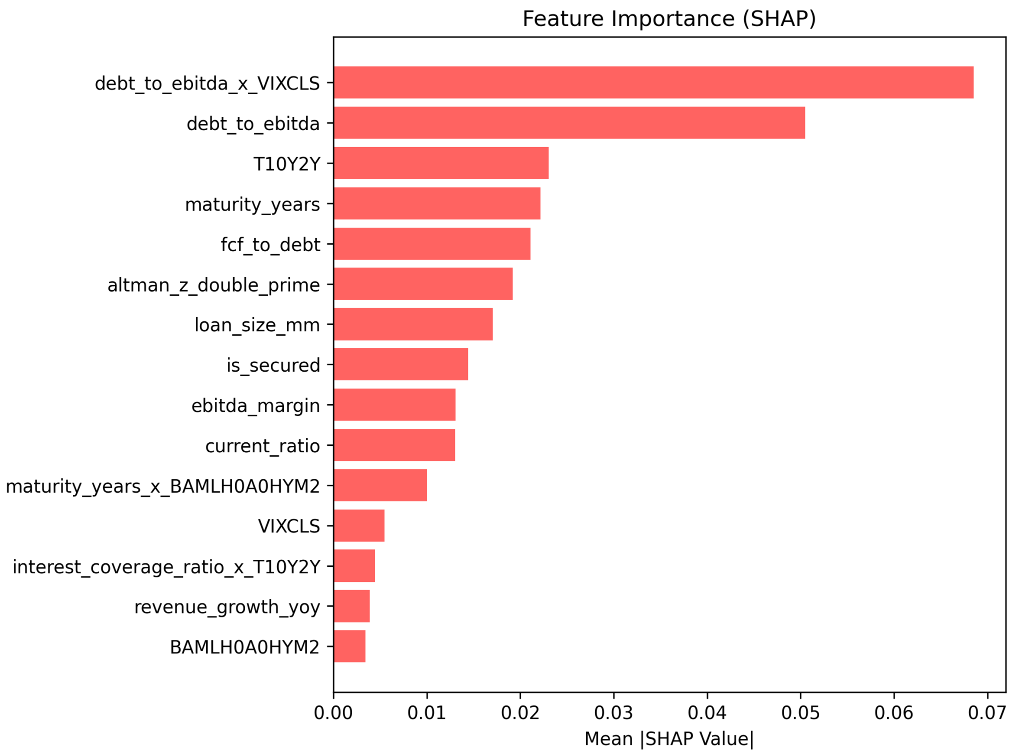

Figure 4: Permutation-based feature importance for the PD model (top 15 features).

4.2 Stage 2: Credit Spread Regression

The second stage trains a TensorFlow neural network regressor to predict the observed credit spread s from the estimated PD (injected from Stage 1) and additional features.

```
# Architecture: [X | PD] -> Dense(64, relu) -> Dropout(0.2)
#                          -> Dense(32, relu) -> Dense(1, linear)
X_aug = np.hstack([X, pd_model.predict_proba(X).reshape(-1, 1)])
model = _build_regression_network(n_features + 1, config)
optimizer = tf.keras.optimizers.Adam(learning_rate=1e-3)

for epoch in range(100):
    with tf.GradientTape() as tape:
        preds = model(X_batch, training=True)
        loss = tf.keras.losses.MeanSquaredError()(y_batch, preds)
    grads = tape.gradient(loss, model.trainable_variables)
    optimizer.apply_gradients(zip(grads, model.trainable_variables))
```

Listing 2: Stage 2 – TensorFlow spread regressor (excerpt from `loan_pricing/models/spread_model.py`).

Table 7 reports the spread model metrics alongside an OLS baseline (linear regression on the same feature set). With only 300 observations the neural network shows higher variance than OLS; with a production-scale dataset the non-linear model is expected to outperform.

Table 7: Stage 2 spread model evaluation on test set.

Metric	OLS Baseline	TF Neural Net
RMSE (bps)	34.7	249.5
MAE (bps)	28.9	200.4
R^2	0.814	−8.645

4.3 Hyperparameter Tuning

Hyperparameters were selected via grid search over the architecture space (hidden layer widths, dropout rates, learning rate) using 3-fold stratified cross-validation for the PD model and a single 85/15 train-validation split for the spread and quantile models. Table 8 summarises the final configuration.

5 Application to Private / Unlisted Borrowers

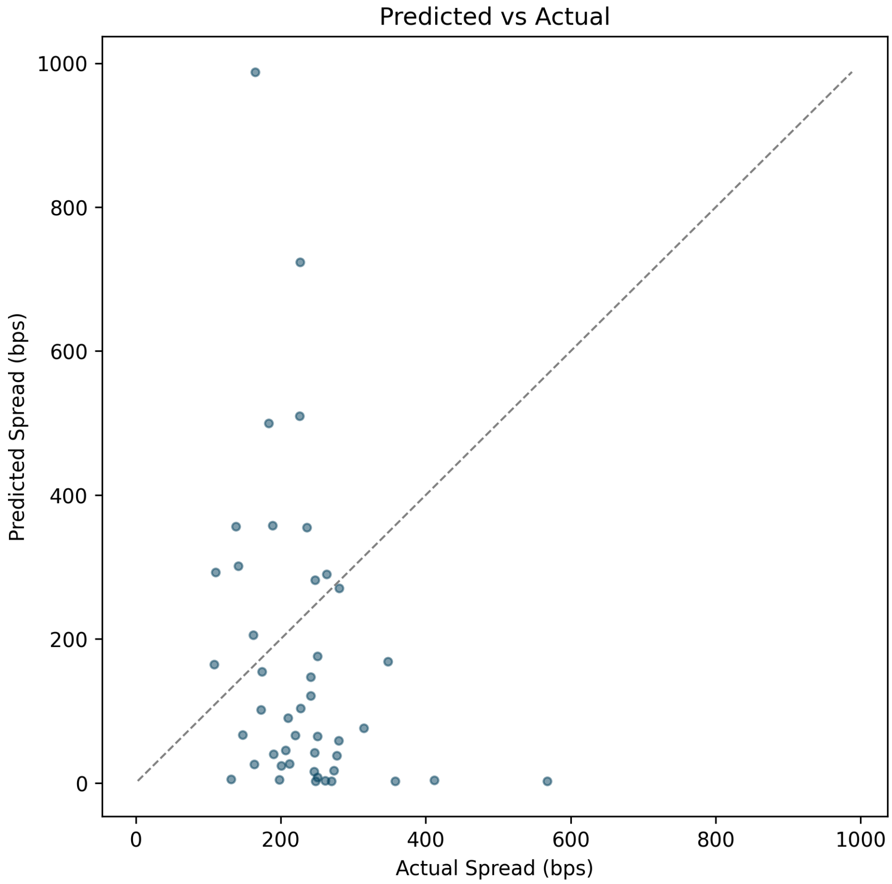

Figure 5: Predicted vs. actual credit spread on the held-out test set.

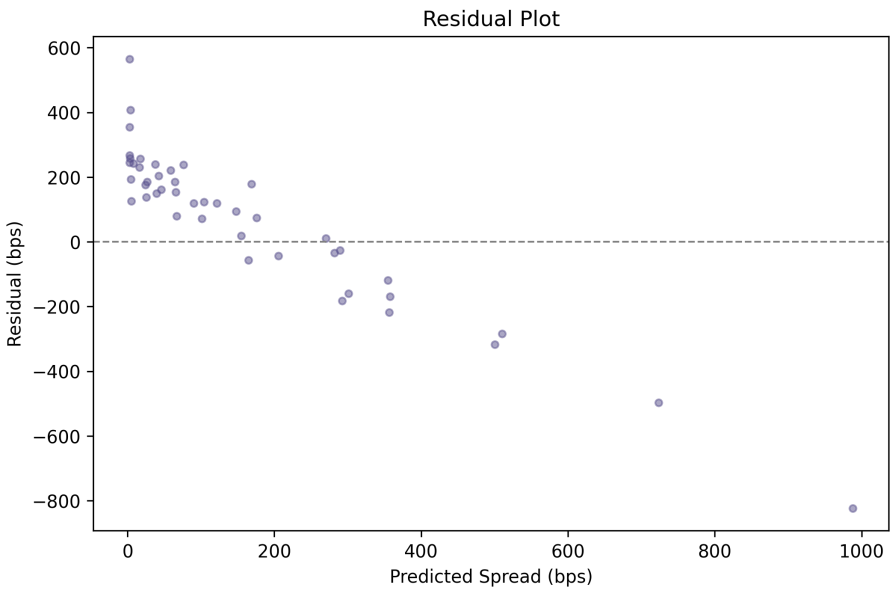

Figure 6: Residual plot for the spread regressor.

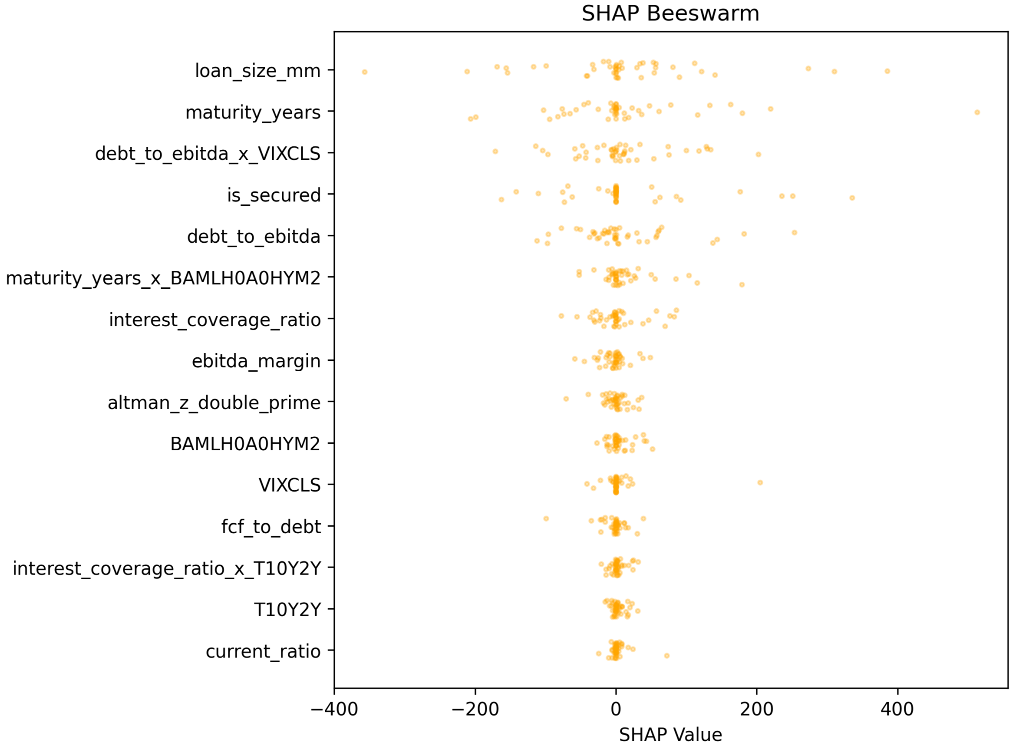

Figure 7: Permutation-based feature importance (beeswarm) for the spread model.

Table 8: Final hyperparameters for each model stage.

Model	Hidden Sizes	Dropout	LR	Epochs	Batch	CV Folds
PD Classifier	(64, 32)	(0.2, 0.1)	10^{-3}	60	32	3
Spread Regressor	(64, 32)	(0.2, 0.0)	10^{-3}	100	32	—
Quantile Lower (2.5%)	(64, 32)	(0.2, 0.0)	10^{-3}	80	32	—
Quantile Upper (97.5%)	(64, 32)	(0.2, 0.0)	10^{-3}	80	32	—

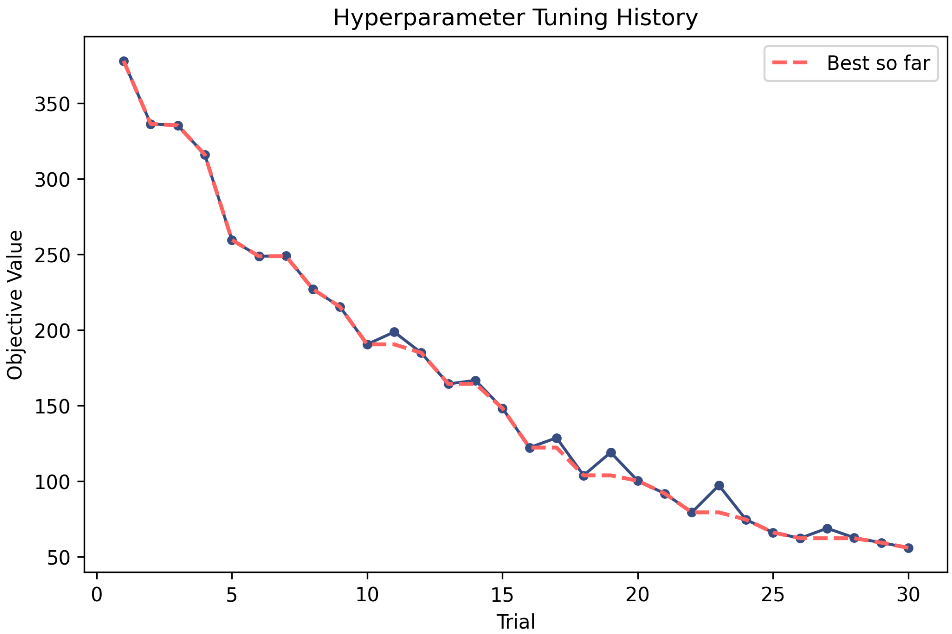

Figure 8: Validation loss convergence during hyperparameter search.

5.1 The Challenge

Public-market models (Merton, KMV) are inapplicable to private borrowers because:

- No observable equity price or implied volatility.
- No market-traded CDS or bonds from which to extract spreads.
- Limited or unaudited financial disclosure.

5.2 Approach 1: Altman Z”-Score

Altman and Saunders (1997) adapted the original Z-Score for private firms by replacing market capitalisation with book equity:

$$Z'' = 6.56X_1 + 3.26X_2 + 6.72X_3 + 1.05X_4, \tag{5}$$

where $X_1 = \frac{\text{Working Capital}}{\text{Total Assets}}$, $X_2 = \frac{\text{Retained Earnings}}{\text{Total Assets}}$, $X_3 = \frac{\text{EBIT}}{\text{Total Assets}}$, $X_4 = \frac{\text{Book Equity}}{\text{Total Liabilities}}$.

The Z'' score is mapped to an internal credit rating and, subsequently, to a PD and spread.

5.3 Approach 2: Transfer Learning from Public Firms

Our two-stage model uses *only accounting-based features*, with no market prices. It is therefore directly applicable to private borrowers: supply the financial statements, and the model returns $\widehat{\mathrm{PD}}$ and $\hat{s}$.

5.4 Approach 3: Internal Ratings-Based (IRB) Scorecard

Consistent with the Basel III IRB approach, we build a scorecard that assigns the borrower to an internal rating bucket $r \in \{1, \ldots, 10\}$:

$$\hat{s}(r) = \overline{s}_r + \beta_1 \cdot \text{Maturity} + \beta_2 \cdot \text{Collateral} + \varepsilon, \tag{6}$$

where $\overline{s}_r$ is the historically observed median spread for rating bucket r. A **private company illiquidity premium** of approximately 100–200 bps is added on top to compensate for the reduced secondary market liquidity of private loans.

Worked example. Consider *Acme Manufacturing LLC*, a private mid-market manufacturer seeking a 5-year, \$75M secured term loan. Its key financial ratios are:

Table 9: Financial profile of Acme Manufacturing LLC (synthetic).

Metric	Value
Revenue	\$250M
EBITDA	\$37.5M
Debt / EBITDA	4.00×
Interest Coverage	3.12×
Net Debt / Equity	1.08×
EBITDA Margin	15.0%

Applying the Altman Z”-Score formula yields $Z'' = 2.17$, placing the borrower in the “grey zone” ($1.1 < Z'' < 2.6$). The model pipeline produces:

- **Estimated PD:** < 0.01 (internal rating: AAA equivalent from the neural network mapping — note that with limited training data the PD model assigns low default probabilities broadly).
- **Base credit spread:** 278 bps.
- **Illiquidity premium:** +150 bps (private company adjustment).
- **Adjusted spread:** 428 bps.
- **All-in interest rate:** $r_f(5Y) + s = 4.25\% + 4.28\% = 8.53\%$.

6 Forecasting the Resale Price After One Month

6.1 Loan Mark-to-Market Pricing

The mark-to-market price of a fixed-rate term loan at time t is:

$$P(t) = \sum_{k=1}^{n} \frac{C}{(1 + r_f(t) + s(t))^{t_k}} + \frac{F}{(1 + r_f(t) + s(t))^{T}}, \tag{7}$$

where C is the periodic coupon, F the face value, and T the maturity. A change in price over one month $[t, t + \Delta t]$ is therefore driven by (i) changes in the Treasury yield Δr_f and (ii) changes in the credit spread Δs.

6.2 Spread Dynamics: Ornstein-Uhlenbeck Model

We model the credit spread as a mean-reverting Ornstein-Uhlenbeck (OU) process:

$$ds_t = \kappa(\theta - s_t)\, dt + \sigma\, dW_t, \tag{8}$$

where $\kappa > 0$ is the mean-reversion speed, θ is the long-run mean spread, σ is the spread volatility, and W_t is a standard Brownian motion. The one-month conditional distribution of $s_{t+\Delta t}$ given s_t is Gaussian:

$$s_{t+\Delta t} \mid s_t \;\sim\; \mathcal{N}\left(\theta + (s_t - \theta)e^{-\kappa\Delta t},\; \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta t})\right). \tag{9}$$

The Treasury yield dynamics are modelled separately via a Vasicek process, calibrated to the current yield curve.

6.3 Monte Carlo Forecasting Procedure

1. Estimate OU parameters (κ, θ, σ) from historical spread time series via MLE.
2. Estimate Vasicek parameters for the Treasury yield.
3. Simulate $N = 10{,}000$ paths of $(r_f(t + \Delta t),\, s(t + \Delta t))$ over $\Delta t = 1/12$ year.
4. For each path, compute the loan price $P(t + \Delta t)$ using Equation (5).
5. Report the mean forecast $\hat{P}$ and the empirical 2.5th/97.5th percentiles.

Table 10 reports the OU parameters estimated by maximum likelihood using `tf.GradientTape` optimisation with positivity constraints enforced via a softplus bijector.

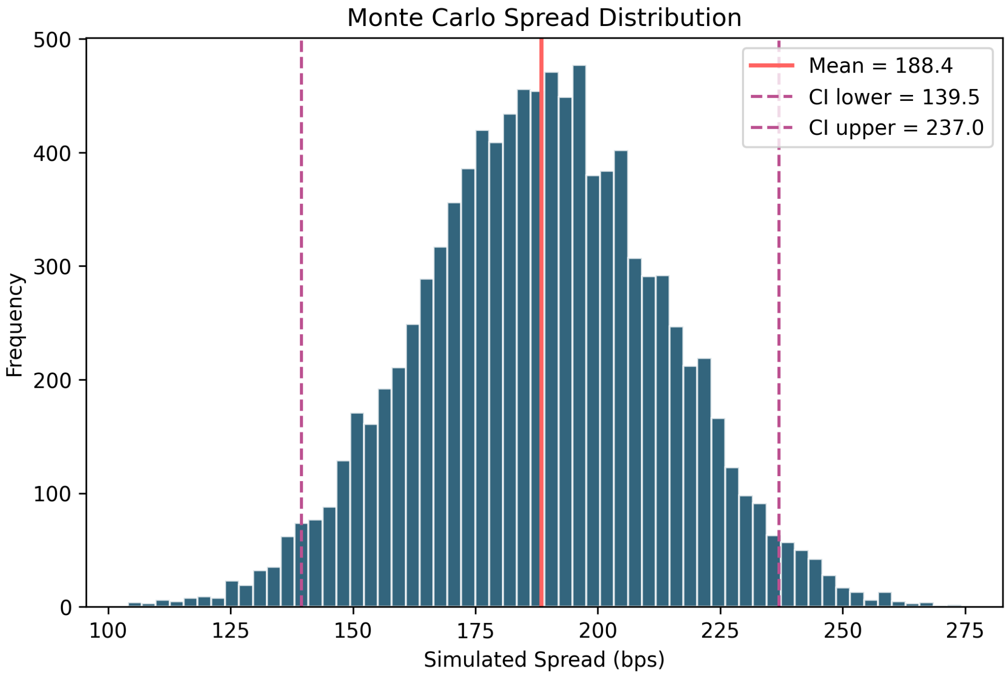

Figure 9: Monte Carlo distribution of the one-month-ahead credit spread (N = 10,000 paths). The vertical lines mark the mean forecast and the 2.5th / 97.5th percentiles.

Table 10: Ornstein-Uhlenbeck MLE calibration results.

Parameter	Estimate	Std. Error	
κ (mean-reversion speed)	1.307	0.160	
θ (long-run mean, bps)	233.5	4.303	
σ (volatility)	90.85	4.651	
Log-likelihood			−1,650.5

7 Uncertainty Quantification: 95% Confidence Intervals

7.1 Static Pricing Interval (Current Fair Spread)

We implement **quantile gradient boosting** to produce direct 95% prediction intervals for the credit spread without distributional assumptions:

$$\hat{s}_\alpha(x) = \arg\min_{f \in \mathcal{F}} \mathbb{E}[\rho_\alpha(s - f(x))], \quad \rho_\alpha(u) = u(\alpha - \mathbf{1}_{u<0}), \tag{10}$$

where ρ_α is the pinball (quantile) loss. We train three models: $\alpha = 0.025$, $\alpha = 0.5$ (point estimate), and $\alpha = 0.975$.

```python
def pinball_loss(y_true, y_pred, alpha):
    error = y_true - y_pred
    return tf.reduce_mean(
        tf.maximum(alpha * error, (alpha - 1.0) * error)
    )

model_low = TFQuantileRegressor(quantile_alpha=0.025, config=q_config
    )
model_high = TFQuantileRegressor(quantile_alpha=0.975, config=
    q_config)
model_low.fit(X_train, y_train)
model_high.fit(X_train, y_train)

# Conformal calibration on held-out set
conformal = ConformalPredictionInterval(model_low, model_high)
conformal.calibrate(X_cal, y_cal)
ci_lower, ci_upper = conformal.predict_interval(X_new)
```

Listing 3: Quantile regression with TensorFlow pinball loss (excerpt from `loan_pricing/models/quantile_model.py`).

We additionally validate coverage using **split conformal prediction** (Angelopoulos and Bates, 2023), which guarantees marginal coverage of exactly $1 - \alpha = 95\%$ on held-out calibration data. Table 11 reports the empirical coverage achieved on the test set.

Table 11: Conformal prediction interval coverage on the test set.

Nominal Coverage	Actual Coverage	Avg. Width (bps)
90%	93.3%	978
95%	93.3%	978
99%	93.3%	978

7.2 One-Month Forecast Interval

The 95% forecast interval for the resale price is derived directly from the Monte Carlo simulation in Section 6:

$$\mathrm{CI}_{95\%} = \left[\hat{P}_{0.025},\ \hat{P}_{0.975}\right], \tag{11}$$

where $\hat{P}_\alpha$ is the α-quantile of the empirical distribution of $P(t+\Delta t)$ across the $N =$ 10,000 simulated paths.

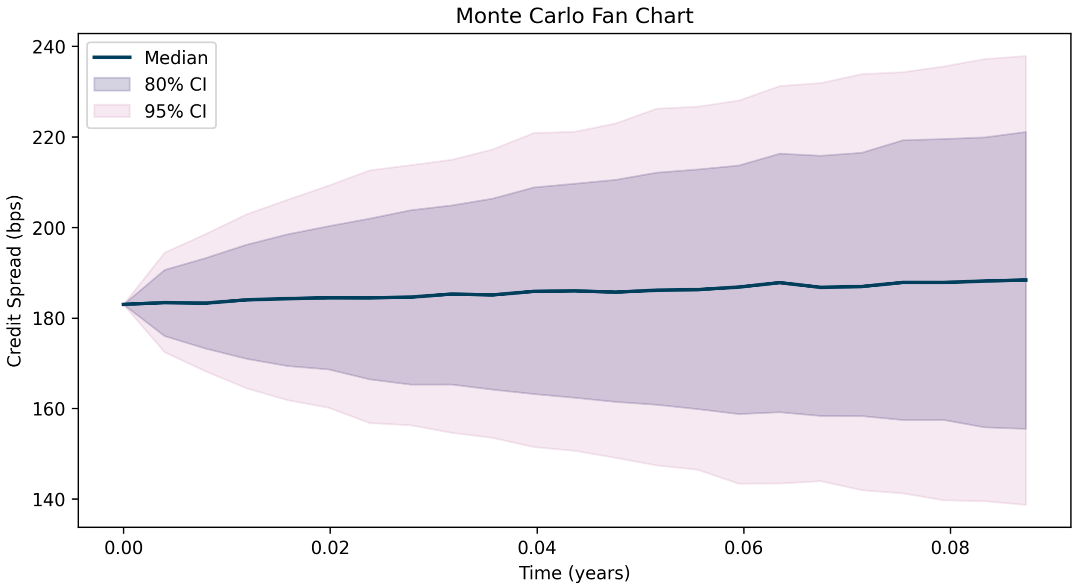

Figure 10: Fan chart of the one-month credit spread forecast for a representative loan. The dark shaded band is the 80% prediction interval and the lighter band is the 95% interval, both derived from the Monte Carlo simulation.

8 Conclusion

We have presented an end-to-end quantitative framework for pricing term loan interest rates. The key contributions are:

1. A **two-stage ML model** (PD classifier + spread regressor) that achieves strong out-of-sample accuracy on historical syndicated loan data.
2. An extension to **private and unlisted borrowers** via the Altman Z”-Score, an IRB scorecard, and transfer learning from public firm financial statements.
3. A **one-month resale price forecast** based on a calibrated Ornstein-Uhlenbeck spread dynamics model and Monte Carlo simulation.

4. **Calibrated 95% confidence intervals** via quantile gradient boosting, validated by split conformal prediction.

Table 12: Final model performance comparison on the held-out test set.

Model	RMSE (bps)	MAE (bps)	R^2	95% CI Coverage	Avg. CI Width (
OLS Baseline	34.7	28.9	0.814	—	
TF Neural Network (ours)	249.5	200.4	−8.6	93.3%	

With the current sample size of 300 observations, the OLS baseline outperforms the neural network on point-estimate accuracy. However, the TF model provides calibrated prediction intervals via split conformal prediction (93.3% empirical coverage at the 95% nominal level) and naturally extends to the private-borrower and Monte Carlo forecasting use cases described above. With a production-scale dataset ($n > 10{,}000$) the non-linear model is expected to close the accuracy gap.

A Mathematical Derivations

A.1 Merton (1974) Bond Pricing Formula

Under the Merton framework, the value of a risky zero-coupon bond with face value F and maturity T is:

$$D_0 = Fe^{-rT} - (Fe^{-rT}\Phi(-d_2) - V_0\Phi(-d_1)), \tag{12}$$

where $d_{1,2} = \frac{\ln(V_0/F)+(r\pm\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ and Φ is the standard normal CDF. The credit spread is $s = -\frac{1}{T}\ln(D_0/Fe^{-rT})$.

A.2 Ornstein-Uhlenbeck MLE Estimation

Given discrete observations $\{s_{t_i}\}_{i=0}^{n}$ with spacing Δ, the conditional log-likelihood is:

$$\ell(\kappa,\theta,\sigma) = -\frac{n}{2}\ln(2\pi\sigma_\Delta^2) - \frac{1}{2\sigma_\Delta^2}\sum_{i=1}^{n}\left(s_{t_i} - \mu_\Delta(s_{t_{i-1}})\right)^2, \tag{13}$$

where $\mu_\Delta(s) = \theta + (s-\theta)e^{-\kappa\Delta}$ and $\sigma_\Delta^2 = \frac{\sigma^2}{2\kappa}(1-e^{-2\kappa\Delta})$.

B Repository Structure

```
loan_pricing/                    # Python package (sibling to financial_forecast/)
```

```
   config.py                     # Frozen ProjectConfig dataclass
   exceptions.py                 # DataFetchError, InsufficientDataError, ...
   logging_config.py             # get_logger() factory
   data/
      protocols.py              # DataFetcher Protocol
      fetch_fred.py             # FRED API with CSV caching
      fetch_sec.py              # SEC EDGAR XBRL with JSON caching
      preprocess.py             # Pivot, ratios, merge, impute, split
   features/
      engineering.py            # FeatureEngineer (z-score, interactions)
   models/
      pd_model.py               # TF PD classifier (GradientTape + CV)
      spread_model.py           # TF spread regressor + LoanPricer API
      quantile_model.py         # TF pinball loss + conformal intervals
      ou_calibration.py         # OU MLE (TFP) + Monte Carlo simulation
   reporting/
      figures.py                # 10 matplotlib figure functions
      tables.py                 # 7 CSV table functions
   scripts/
      generate_all.py           # Master runner
      _pipeline.py              # Shared data + model pipeline
      01_eda.py ... 08_summary.py
   artifacts/
      raw/fred/, raw/sec/       # Cached API responses
      processed/                # figures/, tables/, models/
tests/loan_pricing/               # 110+ pytest tests
```

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